



EAST WEST UNIVERSITY

**CSE475: Machine Learning
[Fall 2025]**

Related Work

Course Code : CSE475
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Submitted by:

Student Name	Student ID
Rafia Noor Nithin	2022-2-60-120
Fahimah Karim Oishi	2022-2-60-134
Mohammad Ashiquzzaman Hadi	2021-3-60-143
Sahla Fabia Safa	2021-2-60-061

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Related Work

Ref.	Title	Dataset Description	Methods	Results	Pros	Cons	Future Work
[1]	Image-Based Air Quality Analysis Using Deep Convolutional Neural Network	Image-PM _{2.5} (Beijing) — 591 images ($\geq 500 \times 500$) matched with hourly U.S. Embassy PM _{2.5} . Classes: Good < 75, Moderate 75–115, Severe > 115. Dataset: http://www.tour-beijing.com/real_time_weather_photo/	Transfer learning (MatConvNet, ImageNet). Fine-tuned CNN with 3-class softmax & dropout; also 4096-D CNN features → Random Forest (500 trees). Data $\times 4$ by splitting + flipping.	CNN 68.74 % accuracy; RF 63.62%. Qualitative results and prior comparison included	1. End-to-end vision (no explicit transmission/depth). 2. Public dataset + clear setup. 3. Two TL strategies compared	1. 3-class coarse labeling; small N → ~69% acc. 2. Single-city domain.	1. Collect more images across cities/seasons 2. try other CNNs (DAG/RCNN) 3. consider regression or finer bins.
[2]	Ensemble of Deep Neural Networks for Estimating Particulate Matter from Images	Beijing PM Image Dataset — 1,460 outdoor photos ($\geq 584 \times 389$ px) with sky + depth; labels from U.S. Embassy PM _{2.5} data (no public release). Dataset: https://github.com/implicitDeclaration/HVAQ-dataset	Base: VGG-16, Inception-v3, ResNet-50 (ImageNet TL, 1-unit MSE). Meta: FFNN (3 → 32 → 64 → 32 → 1). 3-fold CV; Adam (0.001/0.01); batch 128; ≤ 150 epochs; early stop 10; Augment (h-flip, $\pm 2.5^\circ$ rot, crop 5–30 px $\times$ 10); TTA.	RMSE / R²: VGG-16 60.44 / 0.527 • Inception-v3 64.32 / 0.464 • ResNet-50 53.63 / 0.627 • Linear stack 51.63 / 0.655 • FFNN 49.37 / 0.684 (best) $p < 0.0001$.	1. Non-linear stacking beats single CNNs and simple averages. 2. Reproducible train pipeline + TTA. 3. Demonstrates transfer learning	1. Imbalanced (low-PM dominant) dataset → poor high-PM prediction. 2. Only last FC layer fine-tuned. 3. Dataset not public.	1. Collect more high-PM images. 2. Add weather features (humidity, wind). 3. Test more CNN architectures & multi-modal fusion.

					for visual air-quality tasks.		
[3]	Classification Model of Indoor PM _{2.5} Concentration Using CatBoost Algorithm	Office building IAQ sensors (Beijing): 5-min data (2022-01-18 → 2022-03-29). 14,570 samples after cleaning (PM _{2.5} , RH, CO ₂ , Temp, Light). Binary label >25 µg/m ³ .	CatBoost GBDT (ordered boosting). Inputs: Prev PM, RH, CO ₂ , Temp, Light. Data cleaning (forward fill, RFMICE, 3-σ rule). Split 70/15/15. Optuna + Hyperband HPO (lr, depth, λ). Time-series CV. Baselines: MLP, GBDT, LR, DT, KNN.	CatBoost AUC-ROC 0.949, AUC-PRC 0.928, F1 0.883. Top features: Prev PM > RH > CO ₂ > Temp > Light.	1) Strong accuracy (AUC 0.949) with interpretable SHAP. 2) Time-series CV respects chronology. 3) Practical 5-feature input—easy to deploy with common IAQ sensors. 4) Thorough data cleaning pipeline documented.	1) Single-site, short window (~10 weeks) may limit generalization. 2) Binary label (25 µg/m ³) ignores multi-threshold actions (e.g., 15/35 µg/m ³ daily targets). 3) No explicit external validation across buildings/seasons.	1. Extend to multi-class thresholds (15/25/35 µg/m ³). 2. Multi-site/seasonal validation. 3. Add outdoor PM, ventilation state, occupancy proxies. 4. Tie predictions to closed-loop HVAC control for energy-aware actuation.
[4]	Long-term exposure to PM _{2.5} and its constituents and visual impairment in schoolchildren: A	The study recruited residents from four cities (Shenzhen, Foshan, Maoming, and Zhongshan) in Guangdong province from 2020 to 2021. The selection	Study design & sampling Cross-sectional observational study. exposure (3-yr average) and outcome measured	In this cross-sectional analysis of 59,054 schoolchildren, nearly half (46.3%) had	1. Huge, representative sample 2. High-resolution exposure 3. Detailed outcome measurement	1. Exposure misclassification risk: exposure assigned from home address only 2. Mixtu	1. Validate in other regions with different PM _{2.5} levels 2. Mechanistic studies focusing on BC and OM pathways

	population-based survey in Guangdong province, China	criteria for participants were students who were studying at the chosen school and residing at their current address for over two years. A total of 69,610 children from the selected schools were invited to participate, and 59,054 returned a completed study questionnaire. Children without exposure measurements (n = 247) or visual acuity data (n = 10,309) were excluded, leaving a final study population of 59,054 for the current analysis	around the same period. Population sampling: school-based recruitment from 4 cities; eligibility = attended selected school + lived at current address ≥ 2 years. Outcome measurement (clinical / measurement methods) Visual acuity measured using retro-illuminated logMAR (tumbling-E) charts at standardized distances per national standards. Outcomes used both as a continuous variable (logMAR score) and a binary variable (visual impairment defined as ≤ 4.9). Exposure assessment	visual impairment, and higher long-term exposure to PM_{2.5} and several of its chemical components was consistently associated with greater odds of visual impairment. Comparing the highest to lowest quartiles, adjusted odds ratios for visual impairment were elevated for PM_{2.5} (OR	using standardized logMAR protocol. 4.Mixture methods and sensitivity checks (WQS, QGC, multiple exposure windows, many covariate adjustments)	re methods limited 3.Low pollutant levels in Guangdong	3.Explore interaction with NDVI/green space further
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			(geospatial & remote-sensing methods) Confounding control & variable collection Statistical modeling (main analysis) Weighted Quantile Sum (WQS) regression Quantile G- computation (QGC)	1.23, 95% CI 1.14–1.3 3), OM (OR 1.53, 95% CI 1.40–1.6 7) and BC (OR 1.34, 95% CI 1.26–1.4 3), with OM showing the largest per-IQR effect (OR $\approx$ 1.69). Exposur e–respo nse curves were generall y nonline ar, with associat ions tending to rise at lower concentr ations and plateau ing at higher levels. Mixture analyses			
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				using a weighted quantile sum (WQS) index indicated a positive joint effect (OR per quartile ≈ 1.17) and assigned most of the mixture weight to BC and OM (together explaining the bulk of the signal). Associations were stronger among socioeconomically disadvantaged subgroups (lower household			
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				income, lower parental education) and in areas with less surrounding greenness (lower NDVI), and findings remained robust across multiple sensitivity checks (alternative exposure windows, additional covariate adjustments, exclusion subsets, and Quantile G-computation). These results			
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				point to combustion-related PM _{2.5} fractions (BC and OM) as key contributors to the observed association, though the cross-sectional design and home-address exposure assignment limit causal inference and leave room for residual confounding.			
[5]	VDMS: An Improved Vision Transformer	The study built a comprehensive 1	The VDMS model is a hybrid deep learning	The VDMS model showed	1.High Accuracy & Robustne	1.Limited Extreme-Event Represent	1.Enhanced Data Quality & Coverage:

	r-Based Model for PM2.5 Concentration Prediction	km daily dataset for China (2017–2022) combining multiple data sources: PM2.5 data: Hourly ground measurements from CNEMC aggregated to daily means, supplemented with annual global PM2.5 estimates from ACAG for broader coverage. AOD data: 1 km² products from Gaofen-1 and MODIS MAIAC, preprocessed via radiometric/geometric correction, cloud masking (Fmask), and Gaussian interpolation for Gap-filling. Meteorology: Daily gridded variables (wind, temperature, humidity, pressure) from national weather data. Auxiliary data: NDVI (MODIS,	framework combining spatial, temporal, and self-supervised components for PM2.5 prediction. It uses a Vision Transformer (ViT) to extract spatial features from satellite images, SimCLR for self-supervised pretraining to improve feature robustness, a Double-Layered LSTM (DLSTM) to capture temporal patterns, and a multi-head self-attention mechanism to fuse features and model global dependencies. The integrated design enhances spatial-temporal learning and prediction	high predictive accuracy, achieving $R^2=0.93$, RMSE=4.05 $\mu\text{g}/\text{m}^3$, and MAE = 3.23 $\mu\text{g}/\text{m}^3$, outperforming CNN, ViT, and SimCLR-only baselines. Ablation tests confirmed each component (DLSTM, attention, SimCLR) improved performance. The model accurately reproduced spatial and temporal PM2.5 patterns—highlighting high levels in western China (e.g., Xinjiang), lower	ss 2.Strong Generalization 3.Multi-Source Data Fusion 4.Transformer-Based Self-Attention 5.Domain Knowledge Integration	ation 2.Data Smoothing Effects 3.Incomplete Spatiotemporal Modeling: The model treats spatial patterns as relatively static, missing dynamic interactions where pollution hotspots shift with time or weather changes. 4.High Computational Cost	Incorporate higher-resolution satellite data, expand geographic scope, and include more extreme pollution events to improve model robustness. 2.Advanced Spatiotemporal Modeling: Develop methods to better capture dynamic, time-evolving spatial pollution patterns and interactions between space and time. 3.Improved Representation of Heterogeneity: Account
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		monthly) and DEM (ASTER, 30 m aggregated to approx. 1 km).	accuracy, even with sparse data.	pollution along the southeast coast, and declining PM2.5 in Beijing from 58.1 $\mu\text{g}/\text{m}^3$ (2017) to 30.5 $\mu\text{g}/\text{m}^3$ (2022), with seasonal peaks in winter and spring. Overall, VDMS effectively captures and explains pollution dynamics across space and time.			
[6]	The Impact of Particulate Matter (PM2.5) on Human Retinal Development in hESC-Derived Retinal Organoids	Used a lab-based dataset of human embryonic stem cell derived retinal organoids (hEROs) to model retinal development under PM2.5 exposure. Organoids were treated with	The study used an experimental approach exposing retinal organoids (hEROs) from hESCs to PM2.5 (0–100 $\mu\text{g}/\text{mL}$) from day 18 for up to 3 weeks.	PM2.5 exposure inhibited retinal organoid growth in a dose-dependent manner, reducing neural retina formation, organoid size, and	1.Novelty : First study to directly test PM2.5's effects on human retinal development using organoids .	1.Developmental Stage Limitation : Organoids model only early retinal development, not mature or postnatal retina.	1.Study Later Developmental Stages : Examine more mature retinal organoids or extend exposure duration to assess PM2.5 effects on retinal maturation

		0–100 µg/mL PM2.5 for up to 3 weeks, with biological data collected on tissue formation, size, layer thickness, cell proliferation/apoptosis, and gene expression (RNA-seq) focusing on in vitro toxicological effects rather than environmental measurements.	Researchers assessed effects using microscopy, immunofluorescence, cell proliferation/apoptosis assays, and RNA sequencing to analyze structural, cellular, and molecular changes. This integrated analysis revealed how PM2.5 exposure disrupts retinal development mechanisms	retinal layer thickness. It caused lower cell proliferation, higher apoptosis, and disorganized tissue structure, though key retinal markers remained expressed. RNA-seq revealed 461 altered genes, with disrupted MAPK and PI3K/Akt pathways and reduced FGF8/FGF10 expression, linking PM2.5 exposure to impaired growth and increased cell stress during retinal development.	2.Ethical and Human-Relevant Model: Uses hESC-derived retinal organoids, avoiding ethical issues while providing a realistic human-specific system. 3.Comprehensive Approach: Combines morphological, cellular, and genomic analyses, strengthening causal evidence for toxicity. 4.Physiological Relevance: Organoid	2.High Exposure Doses: The PM2.5 concentrations (up to 100 µg/mL) exceed typical real-world levels, limiting direct health extrapolation. 3.Mixed Cell Populations: Bulk RNA-seq may obscure cell-type-specific effects due to organoid cellular heterogeneity. 4.Lack of In Vivo Factors: Missing systemic elements like immune responses or blood-	and long-term functionality. 2.Establish Realistic Exposure Thresholds: Determine biologically relevant PM2.5 concentrations using in vivo or epidemiological studies to better reflect real-world human exposure. 3.Apply Single-Cell Analysis: Use single-cell RNA sequencing to identify specific retinal cell types most affected by PM2.5, improving precision and interpretability of results. 4.ML approach can automate analysis, reveal hidden biological patterns, and improve predictive
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					s mimic in vivo retinal tissue organization, making findings more applicable to human health. 5.Consistent Multi-Level Evidence : Structural , cellular, and molecular results all align, enhancing robustness and credibility .	retina barrier, which could influence toxicity outcomes . 5.Unquantified Risk Threshold: Cannot define realistic exposure limits for human fetuses or pregnant individuals.	accuracy—turning this experimental framework into a data-driven toxicology model for Understanding and forecasting PM2.5’s impact on human retinal development.
[7]	PM2.5 exposure increases dry eye disease risks through corneal epithelial inflammation and	The 2023 study by Yu et al. used both animal and cell models to study PM2.5 effects on eye health. Male mice were exposed to filtered vs. PM2.5-polluted	The study used both animal and cell experiments to investigate how PM2.5 induces dry eye disease. Mice were exposed to	PM2.5 exposure caused dry-eye-like corneal damage in mice and oxidative stress-driven	Multi-Model Validation & Mechanistic Proof: The study’s use of both	Model & Exposure Limitations: The use of mice and an immortalized cell line limits	1.Human Translation: Conduct clinical or epidemiological studies to confirm PM2.5-related molecular markers and

	mitochondrial dysfunction	air for 3–10 weeks, assessing tear volume, corneal damage, and inflammatory markers. Human corneal epithelial cells (HCE-T) were treated with 40–200 µg/mL PM2.5 for 24–48 hours, measuring cell viability, ROS, mitochondrial function, and gene expression (RNA-seq). Together, the dataset combines physiological and molecular evidence linking PM2.5 exposure to dry eye and oxidative stress.	clean or PM2.5-polluted air for up to 10 weeks, with assessments of corneal damage, tear production, tissue inflammation, and NF-κB/Nrf2 pathway activity. Human corneal cells (HCE-T) were treated with PM2.5 (24–48 h) to measure cell viability, ROS, mitochondrial health, and ATP levels. Mechanistic tests used chemical inhibitors/activators (NAC, Bay 11-7082, tBHQ) and genetic modulation (NF-κB knockdown, Nrf2 overexpression). RNA sequencing identified affected	inflammation in human corneal cells. PM2.5-exposed mice showed corneal epithelial damage, NF-κB activation, increased IL-1β and TNF-α, and suppressed Nrf2/HO-1 antioxidant defense, despite unchanged tear production. PM2.5 reduced cell viability, increased ROS and mitochondrial dysfunction, lowered ATP, and altered 461 genes linked to oxidative stress and inflammation.	animal (in vivo) and cell (in vitro) models provides consistent, complementary evidence. Pharmacological and genetic interventions clearly demonstrated causal roles of the NF-κB and Nrf2 pathways, adding strong mechanistic clarity. Translational Relevance: The work bridges epidemiological findings (PM2.5	direct human relevance, and high PM2.5 doses plus controlled exposure settings may not fully represent typical human pollution exposure patterns.	test protective interventions (e.g., antioxidants). 2. Therapeutic Testing: Explore Nrf2 activators, NF-κB inhibitors, or mitochondrial protectants as potential treatments for pollution-induced dry eye. 3. Long-Term and Broader Eye Studies: Examine chronic or reversible effects of prolonged exposure and expand analysis to other ocular tissues (conjunctiva, retina) for a fuller understanding of PM2.5's impact on eye health.
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			genes and pathways, revealing that PM2.5 triggers oxidative stress and inflammation linked to NF-κB activation and Nrf2 suppression.	Mechanistic tests: Activating Nrf2, inhibiting NF-κB, or scavenging ROS (via tBHQ, Bay 11-7082, NAC, or genetic modulation) protected cells from PM2.5-induced damage. PM2.5 triggers corneal injury through the ROS/Nrf2/NF-κB pathway, leading to mitochondrial dysfunction and inflammation underlying dry eye disease.	linked to dry eye in humans) with experimental evidence, successfully reproducing and explaining the dry eye phenotype - enhancing real-world applicability.		
[8]	PM2.5 Concentration	The study gathered 10,084 outdoor images	The method involves two stages:	The image-based	1.Low-cost and	1.Indirect estimation:	1.Multi-location validation:

	Estimation Based on Image Processing Schemes and Simple Linear Regression	from a camera at the Renwu Monitoring Station in Kaohsiung, Taiwan (May–October 2016), capturing photos every 10 minutes between 7 AM–5 PM. Each image was paired with hourly PM_{2.5} measurements and relative humidity data from official sources. The dataset was split 60% for training and 40% for testing, providing labeled data for developing and evaluating an image-based PM_{2.5} prediction model.	1. Image Processing: The system identifies a region of interest (RoI) most sensitive to haze by comparing clear vs. polluted images. Using Sobel edge detection, Otsu thresholding, and morphological operations, it detects areas with the largest visual changes. The brightest region in the difference map is chosen as the RoI, and its average grayscale value becomes the key image feature. 2. Regression Modeling: A linear regression model is trained to map the RoI's average pixel value to	PM_{2.5} estimator achieved a moderate correlation with true measurements, performing best when using the automatically selected RoI. Full dataset: $R^2 = 0.41$, RMSE = 11.9 $\mu\text{g}/\text{m}^3$, better than using the entire image ($R^2 = 0.11$). After filtering out unreliable images (high humidity $\geq 65\%$ or high PM_{2.5} variance): performance improved to $R^2 = 0.73$ and	simple: Uses only a standard digital camera and basic image processing—no need for specialized sensors or complex models. 2. Efficient and lightweight: Employs a single image feature with linear regression, enabling fast computation on consumer hardware. 3. Automatic RoI selection: Improves	Provides visual-based PM_{2.5} inference, not direct measurement, so it cannot replace traditional monitoring instruments. 2. Environmental sensitivity: Accuracy is affected by rapid pollution changes, high humidity, and unaddressed lighting/weather variations. 3. Limited generalization: Trained on a	Test the method across different sites and camera setups to assess adaptability and generalization. 2. Enhanced RoI algorithm: Refine region detection or combine multiple RoIs for improved haze–pollution correlation. 3. Weather and lighting integration: Account for sunlight, cloud cover, and other atmospheric effects to make predictions more robust and accurate.
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			measured PM2.5 concentrations. Once fitted, this regression equation predicts PM2.5 from new images, linking visual haze intensity directly to pollution levels.	RMSE = 8.7 $\mu\text{g}/\text{m}^3$. Region 1 consistently gave the highest accuracy, confirming the ROI-based approach effectively links image haze intensity with air pollution levels.	accuracy by focusing on the most haze-sensitive region, achieving strong correlation ($R^2 \approx 0.73$) with measured PM _{2.5} after filtering.	single camera location—may need recalibration for other viewpoints, cities, or environments. Mental conditions.	
[9]	PM2.5 concentration estimation based on image processing & linear regression.	Image sets used in Sensors study (photo-based)—image features extracted. Classic approach: compute image-level handcrafted features (e.g., saturation, contrast, haze-related indicators) and map them to PM2.5 with	Feature engineering (color/haze metrics) + simple linear regression.	Demonstrates feasibility: image features correlate with PM2.5; simple models produce baseline estimates.	Low complexity; interpretable; needs small data.	Performance sensitive to lighting/scene; handcrafted features miss complex cues.	Combine with learned features; test on diverse field images; add meteorology

		simple regression					
[10]	Detection and prediction of air pollution using ML models	Beijing PM2.5 (UCI) Simple baseline classification and regression on PM2.5	Binary Logit (BL) and Autoregressive (AR)	ACC = 0.999; MSE = 27.000	Very high accuracy for classification baseline; simple models	No DL; low generalization; no preprocessing	Extend to deep learning frameworks for temporal PM2.5 data
[11]	Image-based Air Quality Analysis using Deep CNN	Custom dataset [23] Estimates PM2.5 and classifies categories using raw environmental images	Deep Convolutional Neural Network (CNN)	Accuracy = 68.74%	Demonstrated feasibility of image-based AQI estimation	Limited dataset; no explainability; low accuracy	Use larger dataset and interpretability methods
[12]	UAV-based PM2.5 Detection Using Multispectral Double-Channel Ensemble		CNN (3 blocks), Attention Block, GWAP, Double-Channel Fusion, Weighted Ensemble	Accuracy: 99.60% , MSE: 0.0010, RMSE: 0.0316, MAE: 0.0011, F1: 0.85, FPS:	High accuracy and robustness, Low parameter count (1.57M)• Works in	1. Green channel noise 2. Hardware cost (multispectral UAV)	Multi-location data expansion, Altitude and environmental variability, Lightweight model optimization,

	Learning (DCEL-PM 2.5)	Collected multispectral aerial dataset, Yunlin City, Taiwan (Dec 2021–Nov 2024). 23,100 images (RGB + 4 multispectral: NIR, RE, R, B). Altitude: 100 m. Proposes a double-channel ensemble learning framework for accurate PM2.5 detection using drone-mounted multispectral cameras, focusing on attention-enhanced feature fusion and GWAP.		13	multiple weather conditions • Real-time on embedded systems	3. Complex training architecture	Real-world deployment
[13]	A deep hybrid learning	Worker behavior image dataset	CNN for spatial feature extraction +	Achieved higher accuracy	Strong feature extraction	Domain-specific; not tested	Extend to environmental monitoring

	model to detect unsafe behavior: Integrating CNN and LSTM	Introduced a hybrid CNN-LSTM architecture to automatically recognize unsafe worker actions.	LSTM for temporal sequence learning	than descriptor-based models for recognizing actions	and temporal pattern handling	for environmental data	or multi-sensor data fusion
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